PROJECT PROGRESS REPORT

PROJECT PROGRESS REPORT
(Reporting No. 08/ 10)
***Reporting Period: From** 04/09/2026 **To** 10/09/2025*

1. Student Information

Student Name	MAN VISHNUBHAI PATEL
Enrollment Number	25IC03CA053
Program/Branch	MCA / ICA
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA (DS/ML) /AI &ML
Mentor Name	Mr. Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Project Title

- OpenCV - Based System for Crack Identification and Width Estimation

4. Progress During the Reporting Period

A. Previous Reporting Review

Planned Task	Status	Remarks
Install & Import Libraries	Completed	Required libraries were successfully installed and imported.
Project Configuration	Completed	Project settings, output directories, and OpenCV parameters were finalized.
Output & Mask Directories	Completed	Required folders for processed images, crack masks, and annotated results were completed.
OpenCV Parameters	Completed	Image-processing parameters were configured and optimized through testing.
Pixel-to-mm Calibration	Completed	Calibration settings were integrated with crack-width estimation.
Image Upload	Completed	Multiple crack-image upload and processing were completed successfully.
Image Preprocessing	Completed	Preprocessing and crack-enhancement operations were implemented and tested.
Crack Segmentation	Completed	Crack regions were successfully segmented from processed images.
Crack Detection	Completed	Crack detection was implemented and validated using test images.
Crack Width Estimation	Completed	Crack-width measurements were generated using the detected crack regions and calibration settings.
Risk Classification	Completed	Risk classification was implemented using the defined crack-measurement thresholds.
Result Visualization	Completed	Annotated images and crack-analysis results were generated successfully.
CSV Report	Completed	CSV reporting was implemented to store crack measurements and analysis results.
ZIP Download	**In Progress**	ZIP packaging and download of annotated images, crack masks, processed images, and CSV reports are being finalized.

B. Work Completed During This Reporting Period

- **Activities/tasks completed** major development tasks and integrated the main system workflow.
- **Modules/features developed** Completed crack detection, width estimation, risk classification, visualization, and CSV report modules.

- **Research or literature work completed** Reviewed recent research papers related to crack detection, image processing, and crack-width measurement.
- **Design/development/testing activities** Tested the system with multiple crack images and improved detection, measurement, and result accuracy.
- **Results or outputs achieved** Successfully generated crack detection results, width measurements, risk classification, visual outputs, and CSV reports. Only ZIP Download is pending (5%).

C. Evidence / Deliverables Produced

Sr. No.	Deliverable / Evidence	Description / Link / Attachment
1	Crack Detection & Width Estimation Output	Sample images showing detected cracks and measured crack width.
2	Risk Classification & Visualization	Annotated results showing crack severity and visual analysis output.
3	CSV Report & Source Code	Generated CSV analysis report and updated project source code.

Google Colab Link:

https://colab.research.google.com/drive/1_fe2jRsdn8JjweJfbfACJxXuQkFeZ204#scrollTo=2wq_Ojcvhjku

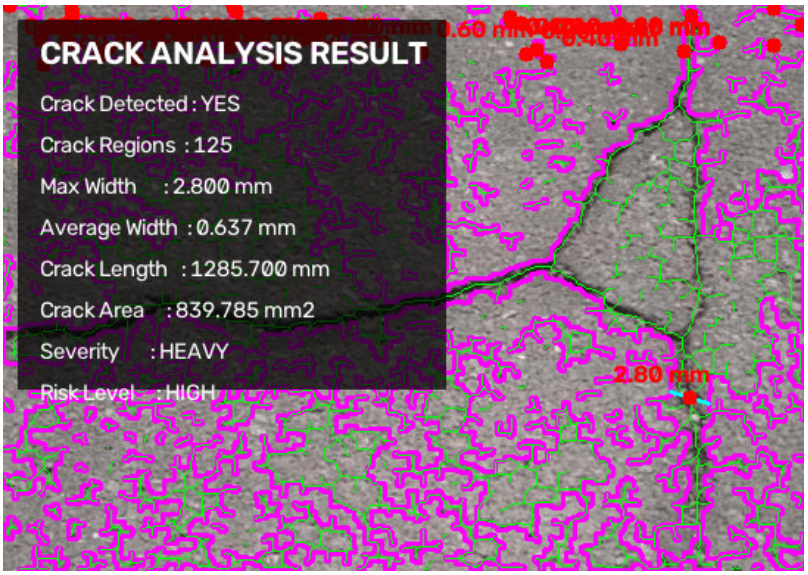
Output:

Zip File Creating:



crack_results.zip

Crack Analysis Result:



D. Current Project Progress

Overall Project Completion: 95%

Sr. No.	Project Component	Status	Completion
1	Library Installation	Completed	100%
2	Import Libraries	Completed	100%
3	Project Configuration	Completed	100%
4	Image Upload	Completed	100%
5	Image Preprocessing	Completed	100%
6	Crack Segmentation	Completed	100%
7	Crack Detection	Completed	100%
8	Crack Width Estimation	Completed	100%
9	Risk Classification	Completed	100%
10	Result Visualization	Completed	100%
11	CSV Report	Completed	100%
12	ZIP Download	**Working**	**90%**

E. Challenges / Issues Faced

- Faced challenges in ZIP Download implementation for combining all generated outputs into a single ZIP file.
- Managing multiple output folders such as images, masks, processed images, and CSV files required proper file handling.
- Ensuring the ZIP file contains the latest generated results without missing or duplicate files.
- Faced minor issues during system testing and integration of the final output modules.
- Required additional testing and debugging to make the ZIP Download feature reliable and user-friendly.

F. Actions Taken / Solution Implemented

- Implemented ZIP file creation to combine all project outputs into one package.
- Organized annotated images, crack masks, processed images, and CSV reports into separate folders.
- Added proper file and folder handling to avoid missing or duplicate files.
- Tested the ZIP generation process with different project outputs.
- Debugged and improved the ZIP Download functionality for reliable final integration.
- Continued testing the remaining ZIP feature, which is currently 90% completed.

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G. Plan for Next Reporting

Sr. No.	Planned Activity	Expected Deliverable	Target Date
1	Complete ZIP Download functionality	Final ZIP package containing images, masks, processed images, and CSV report	17 Sept 2026
2	Perform final system testing and debugging	Tested and error-free complete workflow	17 Sept 2026
3	Finalize documentation and project demonstration	Final report, screenshots, presentation, and demo	24 Sept 2026

H. Learning / Skills Acquired

- **Technical Skills** OpenCV image processing, crack segmentation, detection, width estimation, and pixel-to-mm calibration.
- **Programming Tools** Python, Google Colab, OpenCV, NumPy, Pandas, and Matplotlib.
- **Frameworks** Python-based image processing workflow and Google Colab environment.
- **Research Skills** Research paper reading, literature review, methodology comparison, and technical analysis.
- **Problem-Solving Skills** Debugging, parameter tuning, error handling, and improving system performance.
- **Testing Skills** Testing with multiple crack images and validating detection and measurement results.
- **Documentation Skills** Preparing reports, recording results, creating CSV reports, and maintaining project documentation.
- **Teamwork / Communication Skills** Improved technical communication, progress reporting, project discussion, and presentation skills.

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4. Tasks Accomplished

- Completed Risk Classification and severity assessment.
- Completed Result Visualization with annotated crack images.
- Completed CSV Report Generation for analysis results.
- Tested the complete system using multiple crack images.
- Verified crack detection and width estimation results.
- Improved system performance through debugging and parameter tuning.
- Integrated all completed modules into the final project workflow.
- Started development of the ZIP Download feature, currently at 90% completion.

5. Tasks in Progress / Next Plan

- Complete the ZIP Download functionality for all generated project outputs.
- Integrate ZIP creation with the final project workflow.

- Perform final testing and debugging of the complete system.
- Verify all outputs including crack detection, width estimation, risk classification, visualization, and CSV report.
- Fix any remaining errors and integration issues.
- Complete final documentation, screenshots, and project presentation.

6. Learnings / Skills Acquired

- Improved OpenCV and Python programming skills for image processing.
- Learned practical crack detection, segmentation, and width measurement techniques.
- Gained knowledge of pixel-to-mm calibration and risk classification.
- Improved debugging, testing, and parameter-tuning skills.
- Learned to generate and manage CSV reports and project outputs.
- Improved research, documentation, and technical presentation skills.

References:

- Bodke, K., Bhirud, S., & Sangle, K. K. (2026). Enhanced Crack Width and Depth Measurement through Binary Image Processing and Geometric Analysis. Scientific Visualization, 18(1), 49–61.
- Flah, M., Suleiman, A. R., & Nehdi, M. L. (2020). *Classification and Quantification of Cracks in Concrete Structures Using Deep Learning Image-Based Techniques.* **Cement and Concrete Composites, 114**, 103781.
- Yu, L. (2022). *Intelligent Crack Detection and Quantification in the Concrete Bridge: A Deep Learning-Assisted Image Processing Approach.* **Advances in Civil Engineering**, 2022, 1813821.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks
Progress as per Previous Plan	L3–Apply	All planned tasks completed	Most completed	Partial completion	Very little progress	
Technical / Research Implementation	L4–Analyze	Excellent implementation with clear evidence	Good implementation	Basic implementation	Insufficient evidence	
Quality of Deliverables	L5–Evaluate	High-quality, validated deliverables	Good quality	Acceptable	Poor / incomplete	
Problem Solving & Challenges	L4–Analyze	Challenges clearly identified and effectively solved	Most issues addressed	Limited solutions	No clear effort	
Planning for Next Phase	L6–Create	Clear, realistic and measurable plan	Mostly clear plan	Basic plan	No proper plan	
Documentation & Presentation	L3–Apply	Well-structured and professional	Minor issues	Average clarity	Poor documentation	
Total marks out of 100						

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Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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